

MATLAB 2DOP

• Check DATCOM data if right, drag is 0 & moments show aircraft going down while ~~angle~~ ~~increase~~ AoA increases

Moments

• Moment from the body is 16 & covers pitch angle θ to go up to 100s of thousands while moments from the coefficient cover ~~a~~ a moment of 3 to -3 & output little negative while saturating the q_{bar} (Dynamic pressure) for the ~~any~~ pitch angle

? How does ~~Arctan2~~ ~~Arctan2~~ graph the pitch angle? Is it the value ~~for~~ of angles or is it representing the movement of how the ~~the~~ ~~the~~ aircraft would have moved with the due to pitch angle?

New Day

• Derived ~~magnitude~~ speed is ~~AA~~ $V_x = 201.7 \text{ m/s}$

$$V_2 = 83.061 \text{ m/s}, 56737.3093$$

$$V_2 = 38.4492 \text{ & } V_x = 51.4251$$

Info on any Aircraft p.2. Tail

SAVSI: Sweep angle

CHSTAT: Chord station for sweep reference

SWAFP: Trailing edge sweep parameter

TWISTA: Tail twist angle

SSPND: Cranked / double-delta geometry adjustment

DHDADI: Inner dihedral angle

DHDADO: Outer dihedral angle

Type: Wing / Tail type

TOVC: Thickness-to-chord ratio

XOVC: Maximum thickness location

Tail airfoil

CLI: Ideal lift coefficient

ALPHA1: Zero-lift angle offset

CLALPA: Lift curve slope

CLMAX: maximum lift coefficient

CNO: zero-lift pitching moment coefficient; airfoil pitching tendency

LERI: Leading edge radius parameter

CAMBER: Whether tail airfoil is cambered

CLAMO: Lift coefficient at zero AoA